

Aaron Pacheco - D600 Task 3

A. Describe the purpose of this data analysis by doing the following:

1. Propose **one** research question that is relevant to a real-world organizational situation captured in the provided dataset that can be explored using principal component analysis (PCA) and applied in a regression model to support decision-making.

Can patterns in housing features, neighborhood amenities, and location be used to explain variations in housing prices?

2. Define **one** goal of the data analysis. Ensure that your goal is relevant within the scope of the scenario and is represented in the available data.

The goal of the analysis is to utilize Principal Component Analysis to reduce the number of features among a large number of continuous variables. We can then use those variables in a linear regression to predict housing pricing. This can allow us to identify patterns in housing pricing that can be applied to urban planning and housing development strategies.

B. Explain the reasons for using PCA by doing the following:

1. Explain **one** benefit of using PCA for regression analysis. Include expected outcomes.

One major benefit of using PCA for regression analysis is that it will eliminate the concern of multicollinearity. An issue in housing data is that there are a lot of correlations between data points. (Higher square footage generally means more bedrooms which generally means more bathrooms, etc.) Because of this, you can run into multicollinearity issues in regressions. By utilizing PCA, we should instead see 3-5 principal components with no correlation that we can utilize for regression instead. This should lean to a cleaner regression.

2. Summarize **one** assumption of PCA.

It is assumed that all data used for PCA is continuous, numerical data standardized about a mean of zero. PCA operates on only continuous numerical data and doesn't naturally handle nominal, ordinal, or binary data. PCA assumes data is centered about zero, so it is very important to standardize the data.

C. Summarize the data preparation process for principal component analysis (PCA) by doing the following:

1. Identify the quantitative (continuous) variables that you will need to answer the research question proposed in part A1 to perform principal component analysis (PCA).

SquareFootage
NumBathrooms
NumBedrooms
BackyardSpace

CrimeRate
AgeOfHome
DistanceToCityCenter
EmploymentRate
PropertyTaxRate
PreviousSalePrice

2. Perform the following data cleaning techniques to prepare the data for modeling and provide a screenshot of the completed steps:

- Handling missing values
- Mitigating outliers
- Standardize the continuous variables

Handling missing values:

```
#Drop rows with missing values  
relevant_df=relevant_df.dropna()
```

Mitigating Outliers:

```
#Identify outliers  
col_mean = relevant_df.mean()  
col_std = relevant_df.std()  
  
zscore = (relevant_df-col_mean)/col_std  
  
#Create dataframe without outliers  
clean_df = relevant_df[(zscore.abs()< 3).all(axis=1)]
```

Standardizing Variables:

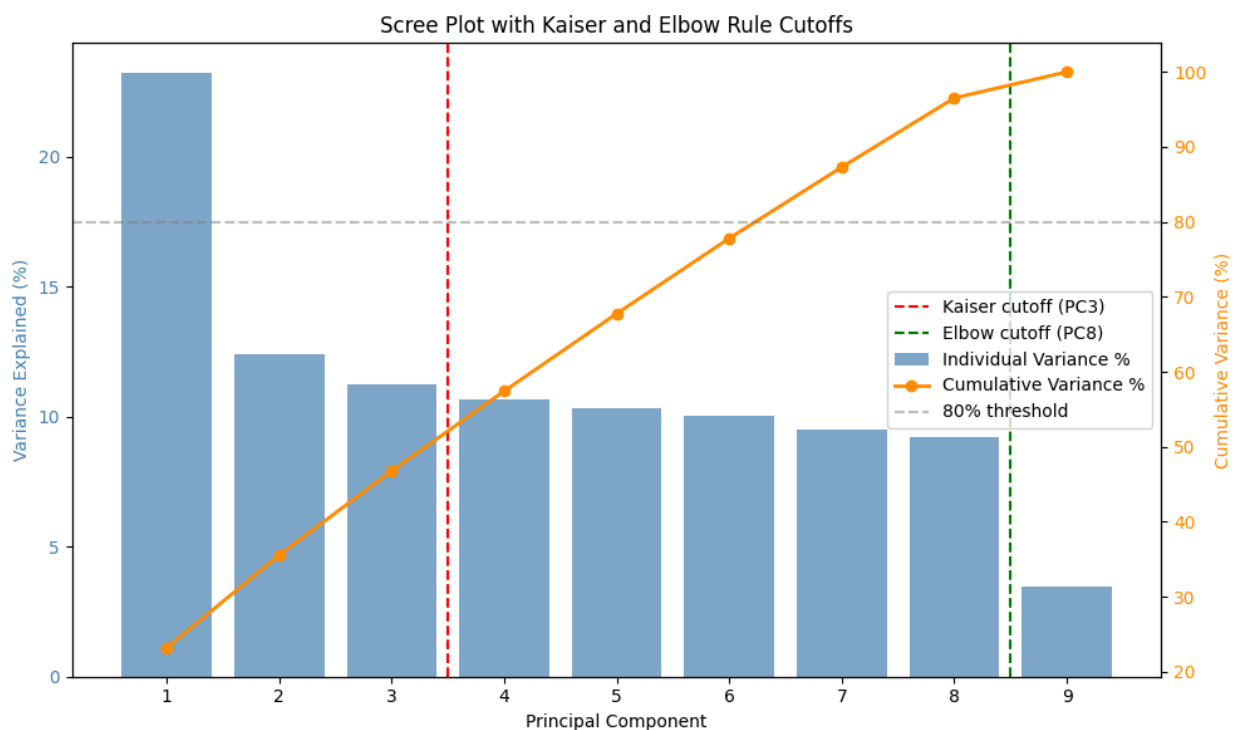
```
#Separate Target from Features  
target = relevant_df['Price']  
features = relevant_df.drop(columns=['Price'])  
  
#Standardize Features  
scaler = StandardScaler()  
scaled_array = scaler.fit_transform(features)  
  
scaled_df = pd.DataFrame(scaled_array, columns=features.columns, index=features.index)  
print(scaled_df.describe().round(2))
```

D. Perform PCA on the prepared dataset from part C2 by doing the following:

1. Provide a screenshot of the principal components' loading matrix.

	PC1	PC2	PC3	PC4	PC5	PC6	PC7	PC8	PC9
SquareFootage	0.480288	0.240281	0.127711	-0.074073	0.047134	-0.023100	-0.566788	0.281863	-0.535109
NumBathrooms	0.410033	0.235533	0.137325	-0.050877	0.042660	0.087671	0.751908	-0.186593	-0.381132
BackyardSpace	0.153643	-0.224599	0.405017	0.860796	0.120931	-0.056537	-0.023372	-0.039815	0.031433
CrimeRate	-0.135158	0.578272	-0.153294	0.316999	-0.384745	0.517845	-0.158144	-0.286012	0.018493
AgeOfHome	-0.212217	0.309221	0.596438	-0.205762	-0.095940	-0.467912	-0.139852	-0.460694	0.036421
DistanceToCityCenter	-0.304380	0.177346	0.424559	-0.008653	-0.325621	0.083103	0.229974	0.727656	0.016389
EmploymentRate	0.123735	-0.523033	0.417808	-0.292230	-0.272831	0.561798	-0.115685	-0.217170	-0.023774
PropertyTaxRate	-0.236170	0.219039	0.237863	-0.099835	0.802182	0.422685	-0.041958	0.030241	0.069752
PreviousSalePrice	0.592075	0.230945	0.100811	-0.112640	-0.011778	0.017113	-0.015443	0.112021	0.748359

2. Identify the total number of principal components that should be retained, using the elbow rule or the Kaiser rule. Include a screenshot of the scree plot.



The Kaiser Rule and the Elbow Rule provide different results for what principal components we should retain. The Kaiser rule suggests we should retain PC1-3 while the Elbow Rule suggests PC1-8 should be retained. I'm choosing to follow the Kaiser rule and only retain PC1-3. When you look at the scree plot, you can see a slight elbow between PC8 and 9, but it isn't exceptionally convincing, so I will choose to utilize the Kaiser rule to utilize a simpler model with less risk of overfitting. This means I am retaining PC1-3.

3. Identify the proportion of variance for each of the principal components identified in part D2.

	Eigenvalue	Proportion of Variance (%)	Cumulative Variance (%)
PC1	2.088	23.19	23.19
PC2	1.114	12.38	35.57
PC3	1.010	11.22	46.79

E. Perform the data analysis and report on the results by doing the following:

1. Create a regression model using the principal components retained from part D2 as predictors in a principal component regression (PCR) model.

```
#Project data onto retained components
n_retained = 3
pca_retained = PCA(n_components=n_retained)
x_pca = pca_retained.fit_transform(scaled_df)

x_pca_df = pd.DataFrame(x_pca,
                        columns=[f'PC{i+1}' for i in range(n_retained)],
                        index=scaled_df.index)

#Fit OLS model on full dataset
x_pca_const = sm.add_constant(x_pca_df)
ols_model = sm.OLS(target, x_pca_const).fit()

print(ols_model.summary())
```

2. Provide a screenshot of the summary of the model or the following extracted model parameters:

- adjusted R^2
- R^2
- probability F statistics
- coefficient estimates
- p -value of each independent variable

OLS Regression Results						
=====						
Dep. Variable:	Price	R-squared:	0.616			
Model:	OLS	Adj. R-squared:	0.616			
Method:	Least Squares	F-statistic:	3748.			
Date:	Fri, 29 May 2026	Prob (F-statistic):	0.00			
Time:	10:48:07	Log-Likelihood:	-90015.			
No. Observations:	7000	AIC:	1.800e+05			
Df Residuals:	6996	BIC:	1.801e+05			
Df Model:	3					
Covariance Type:	nonrobust					
=====						
	coef	std err	t	P> t	[0.025	0.975]

const	3.073e+05	1111.878	276.363	0.000	3.05e+05	3.09e+05
PC1	7.837e+04	769.572	101.830	0.000	7.69e+04	7.99e+04
PC2	2.922e+04	1053.453	27.741	0.000	2.72e+04	3.13e+04
PC3	1.131e+04	1106.475	10.220	0.000	9139.014	1.35e+04

3. Give the root mean squared error (RMSE) of the model.

RMSE: 92999.78

F. Summarize your data analysis by doing the following:

1. List the packages or libraries you have chosen for Python or R and justify how each item supports the analysis.

Pandas - Used to build dataframes from original .csv and PCA data

Numpy - Used to perform mathematical operations required for Kaiser and Elbow rules

Matplotlib.pyplot - Used to build the Scree Plot

Statsmodels.api - Used to perform the Ordinary Least Squares regression using the PCAs

Sklearn.preprocessing (StandardScaler) - Used to scale the data about a mean of zero so PCA could be properly performed

Sklearn.decomposition (PCA) - Used to create Principal Components

2. Discuss the model's adjusted R^2 .

The R^2 value for this model is 0.616. This means that the model explains 61.6% of the variation in housing prices. This tells us that the model is moderately strong at describing housing prices. It is possible that this R^2 value is because of choosing the Kaiser rule instead of the elbow rule. By only using PC1-3 in hopes of eliminating noise, I may have missed additional predictive power in PC 4-8.

3. Discuss the model's RMSE.

The RMSE of the model is \$92,999.78. This tells us that predicted prices deviate from actual prices by \$92,999.78. Knowing the intercept of our model provides us with the predicted price

of an average home of \$307,000, we see that we have approximately 30.3% error. This relatively high RMSE is probably tied to the relatively low R^2 value.

4. Discuss the limitations of your analysis.

My analysis has several limitations that could be contributing to the relatively low R^2 value and the relatively high RMSE. First, housing prices are usually influenced by effects that are non-linear. Secondly, the significant difference in the Kaiser and Elbow tests led me to retaining only PC1-3 in order to avoid additional noise. In hindsight, there may have been significant predictive power in PC4-8 that my analysis avoided due to an unclear elbow in the Scree Plot.

5. Recommend a course of action for the real-world organizational situation from part A1 based on your results.

A course of action worth exploring in relation to our original research question, (*Can patterns in housing features, neighborhood amenities, and location be used to explain variations in housing prices?*), would be to utilize PC1 when screening large property portfolios.

PC1 determines a large portion of the variance in this model and it consists primarily of housing features. It is heavily influenced by factors like SquareFootage and NumBedrooms and PreviousSalePrice. This provides a snapshot into potential home value before taking other factors into consideration. This could be particularly useful when identifying homes as potential flipping targets. If you can identify houses with low PC1 scores but above average prices, this is likely a perfect target for investment. This tells us that the house may be small, but it must be in a very high value location to be fetching the above-average pricing. If one could acquire this property and renovate it properly, it seems very likely that it could earn a lot of value in its location.

F. Acknowledge reference sources used to support the Python or R code application. All references listed should also include an in-text citation in the code annotation. Be sure the sources are reliable. If no sources were used for coding, state, "No sources used."

No Sources Used

G. Acknowledge sources, using in-text citations and references, for content that is quoted, paraphrased, or summarized.

No Sources Used